A.Fill in the blanks in English.(25 points, less than half if in Chinese)

The name of Unix directory “/dev” is from a word 2._____ , “/bin” is from a word 3._____ ; while “src” often means 4._____ .

Command “if -r $FILE” in bash judges whether a file exists and is 5._____ .

While in vi’s insert mode, pressing 6._____ key could help return to command mode.

While linking with static X Window library, “-lX11” means the needed library should be a file called 8._____ .

We met “lazy dog” in our homework task of doing 9._____ practices.

A command “kill -9 3721” means the “kill” program sends a #9 signal to a 10._____ with PID of 3721.

A special macro “$@” inside a makefile rule stands for this rule’s current 11._____ .

In text-based ftp client program, the letter “l” in a command “lcd” likely means 12._____ ; the letter “!” in a command “!pwd” means 13._____ .

The “@” in a Perl variable name “@ages” indicates this variable is of 14._____ type.

15._____ is a program running tutorial for vim; it has 7 lessons for students learning with.

While “.c” is a common suffix for C source code file, “.o” is usually for 16._____ file; and 17._____ is a common suffix for Perl source code file; 18._____ is a common suffix for Python source code file; 19._____ is a common suffix for assembly source code.

Each letter in EDA stands for 20._____ , 21._____ , and 22._____ ; each letter in CAD stands for 23._____ , 24._____ , and 25._____ .

B.Choose single answer from candidate choices.(30 points)

26.'yy4p' in vi command mode means to:

delete 4 lines

duplicate 4 lines

move cursor down 4 lines

move cursor up 4 lines

27.The interpreter program of Tcl/TK scripts:

wish

bash

tk

tcl

28.Interpreter program of Perl scripts:

python

Perl

sh

perl

29.A company was famous for its copying machine products:

Xerox

Intel

Texas Instrument

IBM

30.Key stroke for moving cursor position down a full page in vi:

^f (ctrl-f)

^u (ctrl-u)

^d (ctrl-d)

^b (ctrl-b)

31.It is the world's largest ICCAD company in year 2022:

Cadence

Synopsys

Synopsis

Candence

32.Command can give a Unix command an additional name:

remove

move

alias

rename

33.Command option to tell 'gcc' about position information of header (.h) file:

-h

-H

-L

-I

34.':s/Oct\./Nov./' in vi command line mode performs:

subroutine calling

subtraction

substitution

saving file

35.A Class-B IP address:

127.0.0.1

135.0.1.1

192.168.0.1

10.10.0.21

36.A shell command has different effects than the other 3 commands:

cd

cd ~

cd $HOME

cd $PWD

37.The “-Rt” in a command “ls -Rt” means,

recursively, table orderly

recursively, time orderly

reversely, table orderly

reversely, time orderly

38.A disk partition has its free-space information in:

super block

symbolic bock

data block

inode

39.Command that can NOT display contents of a plain text document:

tail

ls

head

cat

40.Super User of Unix system is also known as:

minister

boss

hacker

root

41.GNU GPL is a technical term most-likely related to:

software license

graphics

debugger

source code complier

42.While using “more” to browse a document, pressing this key could exit the browsing:

x

e

q

b

43.Code version control tool “git” is developed under:

Bill Joy

Steve Wozniak

Bill Gates

Linus Torvalds

44.Unix system time can be changed by command program:

date

sleep

xclock

time

45.Key stroke for moving cursor position up in vi:

^c (crtl-c)

^d (ctrl-d)

j

k

46.It is a network protocol, but NOT on Application Layer:

X Window

TCP

NFS

FTP

47.To repeat last command in shell, type:

$

!!

.

-1

48. Key stroke pair can pause-resume output display of a shell program (where ^ stands for Ctrl key):

^Q-^S

^Z-^C

^C-^Z

^S-^Q

49.Who always listens to a network port and waits to provide network service:

devil

daemon

ghost

agent

50.Honored as inventor of C language:

Richard Stallman

Ken Thompson

Bill Joy

Dennis Ritchie

51.This Unix program can locate a command's position in the file tree:

who

where

whatis

which

52.A C function 'strcasecmp()' is different from another function 'strcmp' in handing:

character number

string length

string boundary

character case

53.When Linux was first mentioned, Linus Torvald was a student in:

U.S.A.

Switzerland

Canada

Finland

54.Output file finally generated by command 'gcc tri.c -lm':

math.h

tri.o

libm.a

a.out

55.To find a certain word inside a file, we can use program:

Find

find

grab

grep

C.Accurately describe each Unix command and its usage in English

Accurately describe the main function and command line syntax of each Unix command in English.

(18 pts, scoring half if in Chinese, but extra explanation in Chinese is very welcome)

56. cut

57. mkdir

58. umount

59. xterm

60. cp

61. top

62. du

63. paste

64. ping

D.Explain the meanings of ALL fields (ALL words and symbols) of the command line in reasonable details; then explain the purpose of the whole command line.

(12 pts, in Chinese OK)

65.find $INC -name std*.h | tee hlist #(~7 fields + whole line)

66.make -f makefile.bit clean; touch demo_bit.c; make -nf makefile.bit #(~7 fields + whole line)

67.export PATH=`pwd`/fake_bin:$PATH; echo $PATH;#(~6 fields + whole line)

68.tar xvf core.tar . ; chmode 764 * #(~8 fields + whole line)

E.Answer questions in reasonable details.

(15 pts, in Chinese OK if not required in English)

69.How to pronounce the project name GNU in English? Why technically we prefer to call our Linux system a GNU/Linux system?

70.Write a shell program 'evenarg5.bash' that prints out 5 times the command line arguments on even number positions (e.g., arg2, arg4, arg6, ···). The program should be able to take unlimited number of arguments. Please describe the program flow first and give enough comments along the code. Magic Symbol is needed too.

71.If you have a Linux computer and a Microsoft Windows computer, an ICCAD software is running on the Linux computer and you want to see its GUI on the Windows computer, what details you should prepare to do?

72.Explain the function of this Perl command line " $str =~ /(([\w._-]+)@([\w._-]+))/ " in detail.

F.附加题

73.试谈一谈如何从各个角度理解集成电路设计工程师需要"expertise in Perl, Tcl, UNIX shell, Makefiles, C, C++ and etc."?

74.How to pronounce the following meta-characters in spoken technical English?

/ \ * @ ({